SEMESTER II | THEORY | 4 CREDITS

Mathematics for Applied Engineering

Comprehensive coverage of Complex Numbers, Vector Algebra, Integral Calculus, Differential Equations, and Statistics for engineering applications.

COURSE CODE	2001D
PROGRAMMES	Chemical, Food Processing, Polymer, Printing, Textile Engineering
CONTACT HOURS	60
CIE / ESE	40 / 60

OFFICIAL SOURCE

Declaration

This handbook is based on the official **SITTTR Kerala Revision 2026** syllabus for **Course 2001D — Mathematics for Applied Engineering**, Semester II.

Verified Source Inconsistencies:

- **CO-PO Matrix:** The PDF contains a garbled articulation matrix (e.g., `C013----------`, `C0112----------`). The exact mapping for each Program Outcome (PO1–PO12) could not be reliably transcribed. This handbook presents the matrix as a best-effort interpretation based on the visible fragments, acknowledging that the official source text is corrupted.
- **Assessment Table:** The PDF lists CIE components (Attendance, CA1–CA6) with individual marks that sum to 165 when added directly, but the official CIE total is 40. This handbook reproduces the official CIE/ESE split (40/60) and the component list as stated, while noting the arithmetic inconsistency.
- **Typo:** Page 2 lists “Measures of depression” in Module IV; this is a typo for “Measures of dispersion” (as correctly written on Page 3). The handbook uses the correct term.

All other content—module topics, subtopics, outcomes, and hours—has been accurately extracted from the PDF and is presented in full depth.

AT A GLANCE

Course Dashboard

📚 Teaching Scheme

L:T:P:S = **3:1:0:0**
Contact Hours: **60**

🎓 Credits & Assessment

Credits: **4**
CIE: **40** · ESE: **60**

🧑‍🏫 Programmes

Chemical · Food Processing · Polymer · Printing · Textile

Pre-requisites

Required prior knowledge

Topic	Course Code	Course Title	Semester
Trigonometry, Determinants, Differentiation	1002	Fundamentals of Engineering Mathematics	I

Teaching & Assessment Scheme

Official scheme (as per syllabus)

Teaching Scheme / Week	Self-learning (SS) / Week	Total Hours / Semester	Credits	Examination Scheme (CIE)	ESE	Total
L:3 T:1 P:0 S:0	As assigned (see Activities)	60	4	40	60	100

Note: CIE components include Attendance, Self-learning activities (Modules 1–4), and two Series Exams. ESE is a written theory exam of 2.5 hours.

STUDY GUIDE

How to Use This Handbook

📖 **Understand**
Read each topic in depth, following the worked examples.
→

📐 **Visualise**
Study the diagrams and use the interactive calculators.
→

✍️ **Apply**
Solve the practice questions and review the model paper.
→

✅ **Revise**
Use the Formula Bank, Quick Revision cards, and Glossary.

LEARNING GOALS

Course Objectives & Outcomes

Course Objectives

1. To provide comprehensive introductory-level coverage of complex numbers.
2. To familiarize students with the basics of vector algebra.
3. To introduce the fundamentals of integral calculus and differential equations for modelling and solving engineering-related problems.
4. To develop an understanding of statistics for data analysis and decision-making in engineering contexts.

Course Outcomes (CO)

CO mapping to Bloom's Taxonomy

CO	Course Outcome	Bloom's Taxonomy Level
CO1	Apply concepts of complex numbers to solve problems.	Apply
CO2	Identify and apply vector algebra techniques to solve problems involving scalar and vector quantities in engineering contexts.	Apply
CO3	Use integration techniques and differential equations to solve basic engineering problems.	Apply
CO4	Apply statistical methods to analyse and interpret data related to engineering and real-life situations.	Apply

CO–PO Mapping

Articulation Matrix (1=Low, 2=Medium, 3=High)

CO \ PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1	-	-	-	-	-	-	-	-	-	-
CO2	3	1	-	-	-	-	-	-	-	-	-	-
CO3	3	1	-	-	-	-	-	-	-	-	-	-
CO4	3	1	-	-	-	-	-	-	-	-	-	-

Source note: The PDF's CO-PO matrix is garbled. The table above is a best-effort interpretation based on fragments (`C013----------`). Correlations are shown as high (3) for PO1 and low (1) for PO2 based on typical engineering mathematics mappings, but please consult the official SITTTR document for definitive mapping.

SYLLABUS MAP

Curriculum Coverage

Module-wise topics and instructional hours

Module	Title	Key Topics	Hours (L:T)	CO
I	Complex Numbers	System, Operations, Modulus & Amplitude, Graphical representation, Polar form	11:4 (15)	CO1
II	Vector Algebra	Scalars & Vectors, Dot & Cross product, Applications (Area, Work, Moment)	12:3 (15)	CO2
III	Integral Calculus & Differential Equations	Integration (Substitution, Parts, Definite), Differential Equations (Variable Separable, Linear)	11:4 (15)	CO3
IV	Statistics	Data classification, Frequency distribution, Measures of Central Tendency & Dispersion	11:4 (15)	CO4

Total Contact Hours: 60 (L:45, T:15). **Self-Learning Hours:** 60 (as detailed in Activities).

CONCEPT FLOW

Module Roadmap

🔢 **Module I**
Complex Numbers
→

📐 **Module II**
Vector Algebra
→

∫ **Module III**
Calculus & DEs
→

📊 **Module IV**
Statistics

Building from foundational numbers to vectors, then to dynamic modelling, and finally to data interpretation.

Module I: Complex Numbers

⏱️ 15 hrs CO1 Apply

Complex numbers extend the real number system to solve equations like $x^2 + 1 = 0$. They are essential in AC circuit analysis, signal processing, and control systems.

The Complex Number System

A **complex number** is defined as $z = x + iy$, where x and y are real numbers, and i is the imaginary unit with the property $i^2 = -1$.

- **Real part:** $\mathrm{Re}(z) = x$
- **Imaginary part:** $\mathrm{Im}(z) = y$

> **Malayalam support:** Complex number-□□ $z = x + iy$ □□□□□ □□□□□□. x □□□ real □□□□, y □□□ imaginary □□□□. i □□□□□□ $\sqrt{(-1)}$ □□□.

> **Example:** For $z = 4 - 3i$, $\mathrm{Re}(z) = 4$, $\mathrm{Im}(z) = -3$.

Conjugate and Equality

Conjugate: The conjugate of $z = x + iy$ is denoted by $\bar{z} = x - iy$.
Equality: Two complex numbers $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$ are equal *iff* $x_1 = x_2$ and $y_1 = y_2$.

> **Example:** If $2x + 3i = 6 - iy$, then $2x = 6 \Rightarrow x = 3$, and $3 = -y \Rightarrow y = -3$.

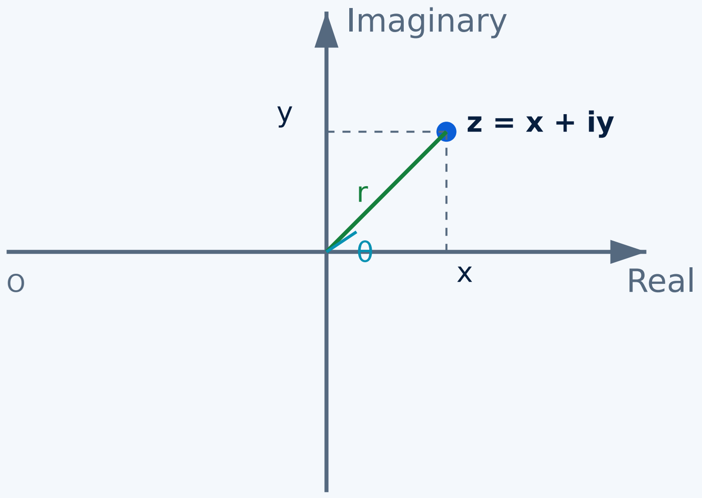
Fundamental Operations

- **Addition:** $(x_1 + iy_1) + (x_2 + iy_2) = (x_1 + x_2) + i(y_1 + y_2)$
- **Subtraction:** $(x_1 + iy_1) - (x_2 + iy_2) = (x_1 - x_2) + i(y_1 - y_2)$
- **Multiplication:** $(x_1 + iy_1)(x_2 + iy_2) = (x_1x_2 - y_1y_2) + i(x_1y_2 + y_1x_2)$
- **Division:** $(1+i)/(1-i) = ((1+i)(1+i))/(1^2+1^2) = i$ (by multiplying numerator and denominator by conjugate of denominator)

> **Example (Division):** Evaluate $(2 + 3i) / (1 - i)$. Multiply by $(1 + i)$: $[(2+3i)(1+i)] / (1^2+1^2) = [(2 - 3) + (2+3)i] / 2 = (-1 + 5i) / 2 = -0.5 + 2.5i$.

Graphical Representation (Argand Plane)

The **Argand plane** is a 2D coordinate system where the x-axis represents the real part and the y-axis represents the imaginary part. Each complex number $z = x + i\,y$ corresponds to a point (x, y).



Argand Plane: point $z = x + i\,y$ with modulus r and amplitude θ.

Modulus and Amplitude

$$|z| = r = \sqrt{(x^2 + y^2)}$$

Modulus (Absolute value)

$$\arg(z) = \theta = \tan^{-1}(y/x)$$

Amplitude (Argument) in degrees

Note: The amplitude is measured from the positive real axis, counterclockwise. For points in different quadrants, add 180° or 360° as needed to get the principal value.

Example: For $z = 3 + 4i$:
$r = \sqrt{(3^2 + 4^2)} = 5$
$\theta = \tan^{-1}(4/3) \approx 53.13°$ (since both x and y are positive, angle is in QI).

🧮 **Complex Number Calculator**

Enter real and imaginary parts to find modulus, amplitude, and polar form.

Real part (x)

3

Imaginary part (y)

4

r = 5.0000, θ = 53.13°, Polar: 5.0000(cos 53.13° + i sin 53.13°)

Polar Form

A complex number can be expressed in **polar form** as:

$$z = r(\cos\theta + i\sin\theta)$$

where $r = |z|$ and $\theta = \arg(z)$

This form is particularly useful for multiplication and division of complex numbers.

Example: Write $z = 1 + i$ in polar form.
$r = \sqrt{1^2 + 1^2} = \sqrt{2}$, $\theta = \tan^{-1}(1/1) = 45°$.
Polar form: $\sqrt{2}(\cos 45° + i \sin 45°)$.

📝 **Module I Summary:** You've learned to represent, manipulate, and visualize complex numbers. These tools are foundational for AC circuit analysis (impedance) and signal processing.

Module II: Vector Algebra

⏱️ 15 hrs | CO2 | Apply

Vectors describe quantities with both magnitude and direction—essential in mechanics, electromagnetism, and structural analysis.

Scalars, Vectors, and Unit Vectors

- **Scalar:** A quantity with only magnitude (e.g., mass, speed, temperature).
- **Vector:** A quantity with both magnitude and direction (e.g., force, velocity, displacement).
- **Unit vectors:** Vectors with magnitude 1, denoted by **i**, **j**, **k** along the x, y, and z axes respectively.
- **Position vector:** A vector representing a point's position relative to the origin: $\mathbf{r} = x\,\mathbf{i} + y\,\mathbf{j} + z\,\mathbf{k}$.

> **Example:** Position vector of point P(2, −3, 1) is $\mathbf{r} = 2\mathbf{i} - 3\mathbf{j} + 1\mathbf{k}$.
> Magnitude $|\mathbf{r}| = \sqrt{(2^2 + (-3)^2 + 1^2)} = \sqrt{14}$.

Vector Operations

- **Addition:** $\mathbf{a} + \mathbf{b} = (a_1 + b_1)\mathbf{i} + (a_2 + b_2)\mathbf{j} + (a_3 + b_3)\mathbf{k}$
- **Subtraction:** $\mathbf{a} - \mathbf{b} = (a_1 - b_1)\mathbf{i} + (a_2 - b_2)\mathbf{j} + (a_3 - b_3)\mathbf{k}$
- **Scalar multiplication:** $c\,\mathbf{a} = (c\,a_1)\mathbf{i} + (c\,a_2)\mathbf{j} + (c\,a_3)\mathbf{k}$

Dot Product (Scalar Product)

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}|\,|\mathbf{b}| \cos\theta = a_1b_1 + a_2b_2 + a_3b_3$$

- Used to find the angle between two vectors: $\cos\theta = (\mathbf{a} \cdot \mathbf{b}) / (|\mathbf{a}|\,|\mathbf{b}|)$
- Vectors are **perpendicular** (orthogonal) if $\mathbf{a} \cdot \mathbf{b} = 0$.

> **Example:** Find the angle between $\mathbf{a} = 2\mathbf{i} + 3\mathbf{j} - \mathbf{k}$ and $\mathbf{b} = \mathbf{i} - 2\mathbf{j} + 2\mathbf{k}$.
> $\mathbf{a} \cdot \mathbf{b} = (2)(1) + (3)(-2) + (-1)(2) = 2 - 6 - 2 = -6$.
> $|\mathbf{a}| = \sqrt{(4+9+1)} = \sqrt{14}$, $|\mathbf{b}| = \sqrt{(1+4+4)} = 3$.
> $\cos\theta = -6 / (3\sqrt{14}) \approx -0.5345 \rightarrow \theta \approx 122.3°$.

Cross Product (Vector Product)

$$\mathbf{a} \times \mathbf{b} = (a_2b_3 - a_3b_2)\mathbf{i} - (a_1b_3 - a_3b_1)\mathbf{j} + (a_1b_2 - a_2b_1)\mathbf{k}$$

- The result is a vector perpendicular to both **a** and **b**.
- Magnitude $|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}|\,|\mathbf{b}| \sin\theta$.
- Vectors are **parallel** if $\mathbf{a} \times \mathbf{b} = 0$.

> **Example:** For $\mathbf{a} = \mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$ and $\mathbf{b} = 2\mathbf{i} - \mathbf{j} + \mathbf{k}$:
> $\mathbf{a} \times \mathbf{b} = (2\cdot1 - 3\cdot(-1))\mathbf{i} - (1\cdot1 - 3\cdot2)\mathbf{j} + (1\cdot(-1) - 2\cdot2)\mathbf{k} = 5\mathbf{i} + 5\mathbf{j} - 5\mathbf{k}$.

Engineering Applications

- **Area of parallelogram:** $|\mathbf{a} \times \mathbf{b}|$
- **Area of triangle:** $\frac{1}{2}\,|\mathbf{a} \times \mathbf{b}|$
- **Work done:** $\mathbf{F} \cdot \mathbf{d}$ (dot product of force and displacement)
- **Moment of a force:** $\mathbf{r} \times \mathbf{F}$ (cross product of position vector and force)

> **Example (Work):** A force $\mathbf{F} = 2\mathbf{i} + 4\mathbf{j}$ N displaces an object by $\mathbf{d} = 3\mathbf{i} - \mathbf{j}$ m. Work done $= \mathbf{F} \cdot \mathbf{d} = 2\cdot3 + 4\cdot(-1) = 6 - 4 = 2$ J.

🧮 Vector Calculator

Compute dot product, cross product, and angle between two vectors (3D).

Vector A: x	y	z	Vector B: x
1	2	3	2

y	z
-1	1

Dot: 3, Cross: (5.00, 5.00, -5.00), Angle: 70.89°

📝 **Module II Summary:** Vectors are powerful tools for describing physical quantities. Master dot and cross products—they are the key to solving mechanics and electromagnetics problems.

Module III: Integral Calculus & Differential Equations

⏱️ 15 hrs · CO3 · Apply

Integration is the reverse of differentiation; differential equations model dynamic systems. Together they form the core of engineering mathematics.

Integration as Reverse of Differentiation

Integration undoes differentiation. If $d/dx\,[F(x)] = f(x)$, then $\int f(x)\,dx = F(x) + C$, where C is the constant of integration.

Standard Integrals

$$\int x^n \, dx = x^{n+1}/(n+1) + C$$

$n \neq -1$

$$\int e^x \, dx = e^x + C$$

$$\int 1/x \, dx = \ln|x| + C$$

$$\int \sin x \, dx = -\cos x + C$$

$$\int \cos x \, dx = \sin x + C$$

$$\int \sec^2 x \, dx = \tan x + C$$

Integration Techniques

- **Constant multiple rule:** $\int c\, f(x)\, dx = c \int f(x)\, dx$
- **Sum/Difference rule:** $\int [f(x) \pm g(x)]\, dx = \int f(x)\, dx \pm \int g(x)\, dx$
- **Substitution:**
  - $\int f(ax + b)\, dx$: let $u = ax + b$, $du = a\, dx$.
  - $\int f'(x)\, f(x)\, dx$: let $u = f(x)$, $du = f'(x)\, dx$.
  - $\int f'(x) / f(x)\, dx = \ln|f(x)| + C$.
- **Integration by parts:** $\int u\, dv = uv - \int v\, du$

> **Example (Substitution):** $\int (2x + 1)^3\, dx$.
> Let $u = 2x + 1$, $du = 2\, dx \Rightarrow dx = du/2$.
> $\int u^3\, du/2 = \frac{1}{2} \cdot u^4/4 + C = u^4/8 + C = (2x + 1)^4/8 + C$.

> **Example (Parts):** $\int x\, e^x\, dx$.
> Let $u = x$, $dv = e^x\, dx \Rightarrow du = dx$, $v = e^x$.
> $\int x\, e^x\, dx = x\, e^x - \int e^x\, dx = x\, e^x - e^x + C$.

Definite Integrals

The definite integral $\int_a^b f(x)\,dx$ gives the net area under the curve from $x = a$ to $x = b$. Evaluated as $F(b) - F(a)$, where F is the antiderivative.

> **Example:** $\int_1^2 (3x^2 + 2x)\,dx$.
> Antiderivative: $x^3 + x^2$.
> At $x = 2$: $8 + 4 = 12$. At $x = 1$: $1 + 1 = 2$. Result $= 12 - 2 = 10$.

Differential Equations

- **Order:** The highest derivative in the equation.
- **Degree:** The power of the highest derivative (when the equation is a polynomial in derivatives).
- **Variable Separable:** $dy/dx = f(x)\,g(y) \to \int 1/g(y)\,dy = \int f(x)\,dx$.
- **Linear First Order:** $dy/dx + P(x)y = Q(x)$.
  Integrating factor: $IF = e^{\wedge}(\int P\,dx)$.
  Solution: $y \cdot IF = \int Q \cdot IF\,dx + C$.

> **Example (Separable):** $dy/dx = x^2 / y$.
> $y\,dy = x^2\,dx \to \int y\,dy = \int x^2\,dx \to y^2/2 = x^3/3 + C$.

> **Example (Linear):** $dy/dx + 2y = e^x$.
> $P = 2$, $Q = e^x$. $IF = e^{\wedge}(\int 2\,dx) = e^{2x}$.
> $y\,e^{2x} = \int e^x\,e^{2x}\,dx = \int e^{3x}\,dx = e^{3x}/3 + C$.
> $y = e^x/3 + C\,e^{-2x}$.

🧮 Definite Integral Calculator (Quadratic)

Evaluate $\int_a^b (px^2 + qx + r)$ dx.

p (x^2 coeff)	q (x coeff)	r (constant)	Lower limit a
3	2	0	1

Upper limit b

2

Result: 10.0000

📝 **Module III Summary:** You can now integrate many functions and solve basic differential equations. These tools are widely used in physics, engineering design, and control systems.

Module IV: Statistics

⏱️ 15 hrs | CO4 | Apply

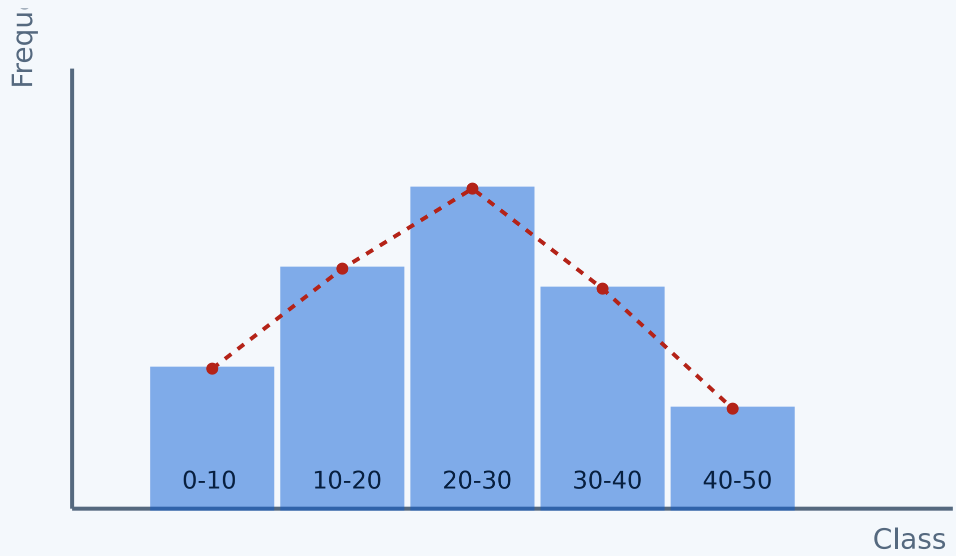
Statistics helps engineers make sense of data, from quality control to process optimization and decision-making.

Data Collection and Classification

- **Primary data:** Collected by the researcher (e.g., surveys, experiments).
- **Secondary data:** Already collected by others (e.g., government records, company reports).
- **Qualitative data:** Categorical (e.g., color, type).
- **Quantitative data:** Numerical (e.g., weight, temperature).
- **Raw data:** Unprocessed data. **Array:** Organized in ascending/descending order.
- **Frequency distribution:** A table showing how often each value or class occurs.

Frequency Distribution & Graphical Representation

- **Class intervals:** Ranges (e.g., 0–10). **Class limits:** The endpoints. **Class boundaries:** Midpoints between upper limit of one class and lower limit of next (e.g., 0–10 becomes −0.5 to 10.5).
- **Histogram:** Bar chart of class intervals vs. frequencies.
- **Frequency polygon:** Line graph connecting midpoints of class tops.
- **Ogive (Cumulative frequency curve):** Plot of cumulative frequency against upper class boundaries.



Histogram with frequency polygon (dashed line).

Measures of Central Tendency

- **Arithmetic Mean:** Sum of values / number of values. For grouped data: $\bar{x} = (\Sigma fm) / \Sigma f$, where m is the midpoint.
- **Median:** Middle value. For grouped data: Median = $L + [(n/2 - cf) / f] \times h$.
- **Mode:** Most frequent value. For grouped data: Mode = $L + [(f_1 - f_0) / (2f_1 - f_0 - f_2)] \times h$.

> **Example (Ungrouped Mean):** Data: 2, 4, 6, 8, 10. Mean = (2+4+6+8+10)/5 = 30/5 = 6.

Measures of Dispersion

- **Mean Deviation:** Average of absolute deviations from the mean.
- **Variance (σ^2):** Average of squared deviations from the mean. For grouped data: $\sigma^2 = [\Sigma f(m - \bar{x})^2] / \Sigma f$.
- **Standard Deviation (σ):** Square root of variance.

> **Example (Ungrouped SD):** Data: 2, 4, 6, 8, 10. Mean = 6. Deviations: $(-4)^2 + (-2)^2 + 0^2 + 2^2 + 4^2 = 16+4+0+4+16 = 40$. Variance = 40/5 = 8. Standard Deviation = $\sqrt{8} \approx 2.83$.

🧮 **Statistics Calculator (Grouped Data)**

Enter class intervals (e.g., "0-10,10-20,20-30") and frequencies (e.g., "5,8,12").

Class Intervals: 0-10,10-20,20-30,30-4(

Frequencies: 4,8,12,6,2

Mean: 23.13, Median: 23.33, Mode: 24.00, SD: 10.73

> 📝 **Module IV Summary:** You can now organize data, visualize it, and compute key statistical measures—essential for quality control and data-driven decisions.

QUICK REFERENCE

Formula Bank

$$|z| = \sqrt{(x^2 + y^2)}$$

Modulus

$$\arg(z) = \tan^{-1}(y/x)$$

Amplitude (degrees)

$$z = r(\cos\theta + i\sin\theta)$$

Polar form

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}|\,|\mathbf{b}| \cos \theta$$

Dot product

$$\mathbf{a} \times \mathbf{b} = |\mathbf{a}|\,|\mathbf{b}| \sin \theta\, \hat{\mathbf{n}}$$

Cross product

$$\int x^n\, dx = x^{n+1}/(n+1) + C$$

Power rule

$$\int u\, dv = uv - \int v\, du$$

Integration by parts

$$dy/dx + Py = Q$$

Linear DE

$$IF = e^{\wedge}(\int P\, dx)$$

Integrating factor

$$\bar{x} = (\Sigma\, fm) / \Sigma\, f$$

Mean (grouped)

$$\sigma = \sqrt{[\Sigma\, f(m-\bar{x})^2 / \Sigma\, f]}$$

Standard deviation

VISUAL SUMMARY

Diagram Library

🔢 Argand Plane

Visualize complex numbers on a 2D plane with real and imaginary axes.

Jump to Module I →

📐 Vector Operations

Understand dot and cross products geometrically.

Jump to Module II →

📊 Histogram & Ogive

Graphical representation of frequency distributions.

Jump to Module IV →

SELF-LEARNING

Suggested Activities

The following activities are recommended for self-study, totalling 60 hours.

Module I (Weeks 1–4)

15 hrs

- Find conjugates, modulus, amplitude, and polar form of complex numbers.
- Solve equality, addition, subtraction problems.
- Multiply and divide complex numbers.
- Explore applications in your engineering field.

Module II (Weeks 5–8)

15 hrs

- Identify scalars/vectors; solve addition/subtraction problems.
- Perform dot and cross products.
- Solve problems on work done and moment.
- Find applications of vectors in your engineering field.

Module III (Weeks 9–12)

15 hrs

- Find integrals of functions.
- Evaluate definite integrals.
- Solve first-order differential equations.
- Find applications of integral calculus in engineering.

Module IV (Weeks 13–15)

15 hrs

- Identify data types and collection methods; represent data graphically.
- Calculate Mean, Median, and Mode for field data.
- Calculate Standard Deviation for small datasets; interpret results.

EXAM PREP

Assessment Methodology

CIE & ESE Scheme

Official CIE and ESE breakdown

Component	Mode	Marks
Attendance	Class attendance	5
CA1	Self-learning activities (Module I)	15
CA2	Self-learning activities (Module II)	15
CA3	Self-learning activities (Module III)	15
CA4	Self-learning activities (Module IV)	15
CA5	Series Exam (2 modules)	15
CA6	Series Exam (2 modules)	15
CIE Total		**40**
ESE	Written theory exam (2.5 hrs)	60
Grand Total		**100**

Note: The PDF's component marks for CA1–CA6 are listed as 15 or 50 in different places, leading to arithmetic inconsistencies. The table above reflects the official CIE total (40) and ESE (60) as stated in the main course description.

Series Examination Pattern (2 hours)

Series test pattern (for 2 modules)

Part	Type	Marks per module	Total (2 modules)
A	2 questions of 1 mark each	2	4
B	3 questions of 3 marks each	9	18
C	2 questions of 7 marks each (with choice)	14	28
Total		**25**	**50**

End Semester Examination Pattern (2.5 hours)

ESE question paper structure

Part	Type	Marks
A	2 questions from each module (1 mark each) — 8 questions total	8
B	2 out of 3 questions from each module (3 marks each) — 8 questions total	24
C	1 set with choice from each module (7 marks each) — 4 questions total	28
Total		**60**

PRACTICE

Module-wise Questions & Answers

Module I: Complex Numbers

Q1: Find the conjugate and modulus of $z = 5 - 12i$. ▲

Conjugate $\bar{z} = 5 + 12i$. Modulus $|z| = \sqrt{5^2 + (-12)^2} = \sqrt{25+144} = \sqrt{169} = 13$.

Q2: Express $z = 1 + \sqrt{3}\, i$ in polar form. ▲

$r = \sqrt{1+3} = 2$. $\theta = \tan^{-1}(\sqrt{3}/1) = 60°$. Polar form: $2(\cos 60° + i \sin 60°)$.

Q3: If $(2+3i)(x+iy) = 1+i$, find x and y. ▲

$(2x-3y) + (2y+3x)i = 1+i$. Equate: $2x-3y=1$, $3x+2y=1$. Solve: multiply first by 2: $4x-6y=2$; second by 3: $9x+6y=3$. Add: $13x=5 \Rightarrow x=5/13$. Then $2(5/13)-3y=1 \Rightarrow 10/13-3y=1 \Rightarrow -3y=3/13 \Rightarrow y=-1/13$.

Module II: Vector Algebra

Q1: Find the unit vector in the direction of a = 2i − 3j + 6k. ▲

$|\mathbf{a}| = \sqrt{4+9+36} = 7$. Unit vector = $(2/7)\mathbf{i} - (3/7)\mathbf{j} + (6/7)\mathbf{k}$.

Q2: Find the angle between a = i + 2j − k and b = 2i − j + 2k. ▲

Dot = $1\cdot2 + 2\cdot(-1) + (-1)\cdot2 = 2-2-2 = -2$. $|\mathbf{a}| = \sqrt{6}$, $|\mathbf{b}| = 3$. $\cos\theta = -2/(3\sqrt{6}) \Rightarrow \theta \approx 105.8°$.

Q3: Compute a × b where a = 2i + 3j + 4k, b = i − 2j + k. ▲

$(3\cdot1 - 4\cdot(-2))\mathbf{i} - (2\cdot1 - 4\cdot1)\mathbf{j} + (2\cdot(-2) - 3\cdot1)\mathbf{k} = 11\mathbf{i} + 2\mathbf{j} - 7\mathbf{k}$.

Module III: Integral Calculus & DEs

Q1: Evaluate $\int (3x^2 + 2x + 1)\, dx$. ▲

$= x^3 + x^2 + x + C$.

Q2: Evaluate $\int_0^1 e^{2x}\, dx$. ▲

Antiderivative = $\frac{1}{2} e^{2x}$. At 1: $\frac{1}{2} e^2$; at 0: ½. Result = $\frac{1}{2}(e^2 - 1)$.

Q3: Solve $dy/dx = 3x^2 y$. ▲

Separate: $dy/y = 3x^2\, dx$. Integrate: $\ln|y| = x^3 + C \Rightarrow y = A\, e$^$(x^3)$.

Module IV: Statistics

Q1: Calculate the mean of the data: 10, 12, 14, 16, 18. ▲

Sum = 70, n = 5, Mean = 14.

Q2: Find the median of: 5, 8, 12, 15, 20, 25. ▲

n=6 (even). Median = average of 3rd and 4th values = (12+15)/2 = 13.5.

Q3: Compute standard deviation for: 2, 4, 6, 8, 10. ▲

Mean = 6. Deviations: $(-4)^2 + (-2)^2 + 0^2 + 2^2 + 4^2 = 40$. Variance = 40/5 = 8. SD = $\sqrt{8} \approx 2.83$.

PRACTICE

Model Question Paper

Based on the official ESE pattern, not an official SITTTR model question paper.

End Semester Examination (2.5 hours)

Course: Mathematics for Applied Engineering (2001D) | **Max Marks:** 60

Part A (8 × 1 = 8 marks)

Answer all questions. Each carries 1 mark.

1. Write the polar form of $z = 2 + 2i$.
2. Find the unit vector of $\mathbf{a} = 3\mathbf{i} + 4\mathbf{k}$.
3. Evaluate $\int 1/x \, dx$.
4. What is the order and degree of $d^2y/dx^2 + 3dy/dx = 0$?
5. Define primary data.
6. What is the median of the data: 3, 7, 9, 12?
7. Write the formula for integration by parts.
8. If $\mathbf{a} \cdot \mathbf{b} = 0$, what can you say about the vectors?

Part B (8 × 3 = 24 marks)

Answer any 2 out of 3 questions from each module. Each carries 3 marks.

Module I: 1. Find the conjugate and modulus of $3-4i$. 2. Express $-1+i\sqrt{3}$ in polar form. 3. Solve $(2+i)z = 3-i$.

Module II: 1. Find the angle between $\mathbf{a} = \mathbf{i}+\mathbf{j}$ and $\mathbf{b} = \mathbf{i}-\mathbf{j}$. 2. Compute the area of parallelogram formed by $\mathbf{a}=\mathbf{i}+2\mathbf{j}$ and $\mathbf{b}=2\mathbf{i}+\mathbf{j}$. 3. Given $\mathbf{F}=2\mathbf{i}+3\mathbf{j}$ and $\mathbf{d}=4\mathbf{i}-\mathbf{j}$, find work done.

Module III: 1. Evaluate $\int 2x(x^2+1)^3 \, dx$. 2. Evaluate $\int_1^2 (3x^2-2x) \, dx$. 3. Solve $dy/dx + 2y = e^x$.

Module IV: 1. Find the mean of grouped data with midpoints 5, 15, 25 and frequencies 3, 5, 2. 2. Compute standard deviation for data: 1, 2, 3, 4, 5. 3. Draw a histogram for the data: 0-10 (5), 10-20 (8), 20-30 (12).

Part C (4 × 7 = 28 marks)

Answer one set from each module (with internal choice). Each carries 7 marks.

Module I: (a) Find the modulus and argument of $z = 4-3i$. (b) Express in polar form and show its position on the Argand plane. *OR* (a) Solve for z: $2z + 3i = 4 - i$. (b) Find conjugate and modulus.

Module II: (a) If $\mathbf{a} = 2\mathbf{i}+3\mathbf{j}-\mathbf{k}$ and $\mathbf{b} = \mathbf{i}-2\mathbf{j}+2\mathbf{k}$, find $\mathbf{a} \cdot \mathbf{b}$ and $\mathbf{a} \times \mathbf{b}$. (b) Find the angle between them. *OR* (a) Find the moment of force $\mathbf{F} = 3\mathbf{i}+\mathbf{j}$ acting at point $\mathbf{r} = 2\mathbf{i}-\mathbf{j}+\mathbf{k}$. (b) Find the area of triangle with sides $\mathbf{a}$ and $\mathbf{b}$.

Module III: (a) Evaluate $\int xe^x \, dx$. (b) Solve $dy/dx + y/x = 1$. *OR* (a) Evaluate $\int_0^{\pi/2} \sin x \, dx$. (b) Solve $dy/dx = y^2/x^2$.

Module IV: (a) Find Mean, Median, and Mode for the data: 0-10 (3), 10-20 (7), 20-30 (10), 30-40 (5). *OR* (a) Calculate standard deviation for the same data. (b) Draw an ogive for the cumulative frequencies.

LAST-MINUTE **Quick Revision**

🔢 Module I

- Complex numbers: $z = x + iy$
- Modulus $r = \sqrt{(x^2+y^2)}$
- Amplitude $\theta = \tan^{-1}(y/x)$
- Polar: $z = r(\cos\theta + \mathrm{i}\sin\theta)$
- Common mistake: forgetting the modulus sign or quadrant for θ

📐 Module II

- Dot: scalar, = 0 for perpendicular
- Cross: vector, = 0 for parallel
- Area of triangle = ½ $|\mathbf{a} \times \mathbf{b}|$
- Work = $\mathbf{F} \cdot \mathbf{d}$
- Common mistake: mixing up dot and cross product formulas

∫ Module III

- Standard integrals: power, exponential, trig
- By parts: $\int u\,dv = uv - \int v\,du$
- Substitution for $f(ax+b)$, $f'(x)f(x)$
- Linear DE: IF = e^($\int$P dx)
- Common mistake: forgetting the constant of integration

📊 Module IV

- Mean: $\Sigma fm/\Sigma f$
- Median: L + [(n/2−cf)/f]×h
- Mode: L + [$(f_1-f_0)/(2f_1-f_0-f_2)$]×h
- SD: $\sqrt{[\Sigma f(m-\bar{x})^2/\Sigma f]}$
- Common mistake: using wrong class boundaries or midpoints

✅ **Exam Checklist:** Know your formulas, practice at least 2 problems per topic, use the calculators to verify answers, and review the model paper structure.

KEY TERMS

Glossary

Important terms and their meanings

Term	Meaning

Term	Meaning
Complex Number	Number of the form $x + iy$
Modulus	Absolute value, distance from origin in Argand plane
Amplitude	Angle made with positive real axis
Scalar	Quantity with magnitude only
Vector	Quantity with magnitude and direction
Dot Product	Scalar product, measures projection
Cross Product	Vector product, gives perpendicular vector
Definite Integral	Integral with specified limits, gives net area
Differential Equation	Equation involving derivatives
Frequency Distribution	Table showing class frequencies
Standard Deviation	Measure of spread from the mean

RESOURCES

References

Textbooks

- SITTTR Kalamasserry — *Mathematics for Applied Engineering*

Web Resources

As per the syllabus, no web resources are officially listed.